

9 (**long**) Figure 6 shows a mechanism in a vertical plane. A pendulum is made of a point mass m attached by a light rod of length l to a pulley of radius $r < l$. An inextensible cable is anchored on the rim of the pulley, and then connected to a second mass m through another pulley. The pulleys are massless and frictionless. The cable and rod are massless. The two masses are subject to gravity. The angle θ is the rotation of the pendulum from vertical.

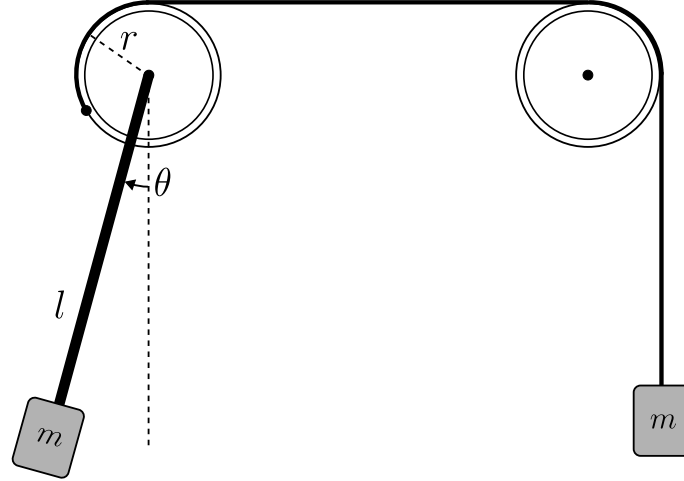


Fig. 6

(a) Find the equilibrium angle θ_{eq} of the pendulum. [7]

(b) The pendulum is displaced by a small angle $\delta\theta$ from its equilibrium position θ_{eq} and released from rest.

(i) Derive the equation of motion of the system by taking moments about the pivot point of the left pulley, and find an expression for the angular frequency of the oscillations. [8]

(ii) Write an expression for the total energy E of the system (to within an arbitrary additive constant). [5]

(iii) Writing the angle θ as $\theta_{eq} + \delta\theta$ and using the fact that $\delta\theta \ll \theta_{eq}$, show that this expression can be approximated as:

$$E = \frac{1}{2}m(r^2 + l^2)\delta\dot{\theta}^2 + \frac{1}{2}mgl \cos(\theta_{eq})\delta\theta^2 + \text{constant}$$

[5]

(iv) By differentiating this expression with respect to time or otherwise, verify your calculation of part (b)(i) for the angular frequency of the oscillations. [5]